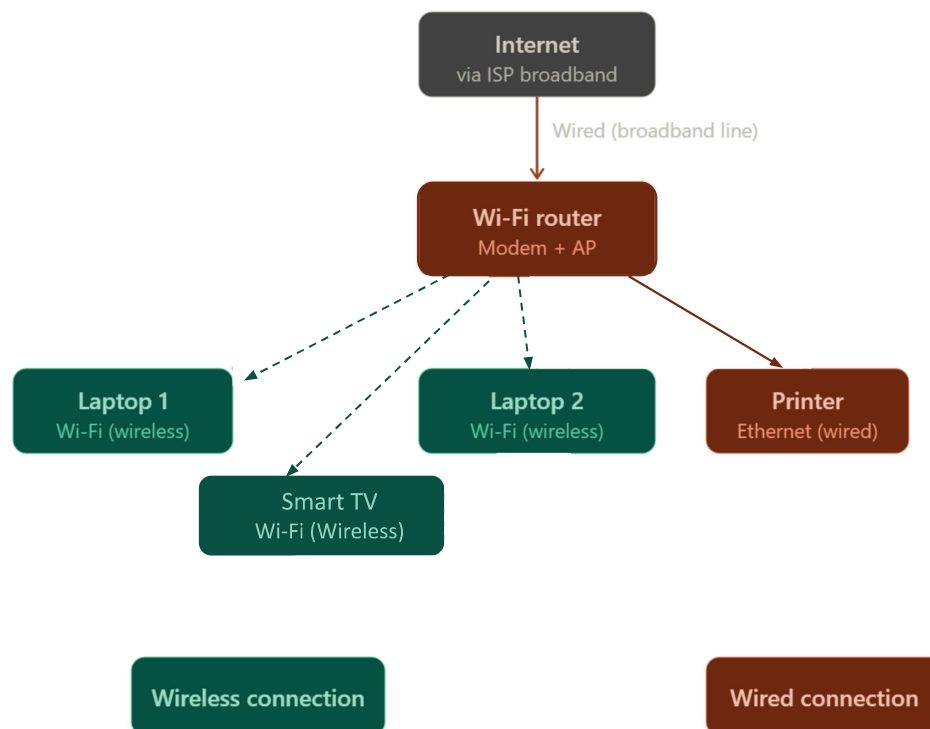


Build a SOHO network

Task 1: Draw a network diagram for a small office/home office (SOHO) setup that includes one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).



Task 2: List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed. Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

Component	Suggested model	Approx. price (₹)	Notes
Broadband connection	Jio Fiber / Airtel Xstream Fiber (ISP-provided ONT)	Plan-based, ONT usually free with subscription	Comes with the ISP's own ONT/modem box
Wi-Fi router	TP-Link Archer C6 (AC1200 dual-band)	₹2,200–2,800	Generally the ISP provider provides a router with built-in Wi-Fi. Else buy a Wi-Fi router that supports WPA2/WPA3
Laptop x2	HP 15s-du2xx + one more laptop	—	
Smart TV	Any Wi-Fi-enabled smart TV	—	
Printer	HP DeskJet 2331 or similar (with Ethernet or Wi-Fi support)	₹3,000–4,500	Pick a model with an Ethernet port if you want it wired.
Ethernet cable (router → printer)	Any Cat6 cable, 1–2 meters	₹100–200	Cat6 is preferred over Cat5e.
Wi-Fi extender (optional)	TP-Link RE200	₹1,200–1,500	Only needed if the signal doesn't reach every room
Switch (optional)	TP-Link LS1005G (5-port Gigabit)	₹700–900	Only needed if you want to wire more than 1–2 devices (e.g., a printer and a desktop).

For your setup specifically (2 laptops + smart TV all wireless, printer wired), you don't strictly need a separate switch—the router's built-in LAN ports can handle the printer directly. A switch becomes necessary only if you outgrow the router's onboard ports.

Task 3: Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

1. Connect a laptop to the router (via Ethernet cable or its default Wi-Fi, printed on the router's label) and open a browser.
2. Go to the router's admin page—typically 192.168.0.1 or 192.168.1.1 (check the sticker on the bottom router).
3. Log in with the default credentials (usually admin/admin, printed on the router or in its manual).
4. Change the default admin password first—go to System Tools and set a new, strong password. This prevents anyone else on the network from taking over the router's settings.
5. Go to Wireless/Wi-Fi settings and set a unique SSID (e.g. HomeOffice_5G)—avoid using your name or address in the SSID.
6. Under Wireless Security, set the security mode to WPA2-Personal (AES)—or WPA3 if the router and your devices support it—and choose a strong Wi-Fi password (12+ characters, a mix of letters/numbers/symbols).
7. Disable WPS (Wi-Fi Protected Setup) if not needed—it's a known weak point that can be exploited to bypass your password.
8. Save/apply settings—the router will usually reboot.
9. Reconnect all devices (laptops, smart TVs) to the new SSID with the new password.
10. Check for the router's firmware update from the admin panel (System Tools → Firmware Upgrade) to patch known security vulnerabilities.

Task 4: Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Device	MAC Address (example)	IP Assignment	IP Address
Wi-Fi Router	AA:BB:CC:00:00:01	— (gateway)	192.168.1.1
Printer	AA:BB:CC:00:00:02	Static	192.168.1.10
Smart TV	AA:BB:CC:00:00:03	Static	192.168.1.11

Laptop 1	AA:BB:CC:00:00:04	DHCP	192.168.1.101
Laptop 2	AA:BB:CC:00:00:05	DHCP	192.168.1.102

To set static IPs on the printer/TV, go into their network settings menu, switch from "Automatic (DHCP)" to "Manual/Static," and enter the IP, subnet mask (255.255.255.0), and gateway (192.168.1.1). It's also worth reserving those addresses in the router's DHCP settings (DHCP reservation by MAC address) so the router never hands them out to another device by mistake.

Task 5: Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Cause 1 — Wrong Wi-Fi password or wrong network selected Diagnosis: Check if the laptop shows "connected, no internet" or fails to connect entirely. Fix: Forget the network on the laptop and reconnect, carefully re-entering the SSID and password. Confirm you're not accidentally connected to a neighbor's similarly-named network.

Cause 2 — IP address conflict or DHCP failure Diagnosis: Run ipconfig in Command Prompt—if the laptop shows an address starting with 169.254.x.x, that means it failed to get an IP from the router's DHCP server. Fix: Run ipconfig /release, then ipconfig /renew. If that doesn't work, restart the router. Also check the router's DHCP settings haven't run out of available addresses in the pool.

Cause 3—Router or ISP outage, not the laptop itself Diagnosis: Check if other devices (the other laptop, phone, and smart TV) also can't get online—if none of them can, the problem is upstream, not the laptop. Fix: Restart the router and modem. If the outage persists, contact the ISP to check for a service outage in the area.